Homework 4 - Theory

$\boxed{1a}$ For valid probability mass function, probabilities must be non negative and sum to 1

$$\theta_1 \geq 0, \quad 2\theta_1 \geq 0, \quad \theta_2 \geq 0$$

Since $2\theta_1 \geq 0$ is implied by $\theta_1 \geq 0$ this reduces to

$$\theta_1 \geq 0, \quad \theta_2 \geq 0$$

Also: $P(x=1) + P(x=2) + P(x=3) = 1$

so: $\theta_1 + 2\theta_1 + \theta_2 = 1 \rightarrow 3\theta_1 + \theta_2 = 1$

Thus the constraint is: $\theta_1 \geq 0, \theta_2 \geq 0, 3\theta_1 + \theta_2 = 1$

$\boxed{1b}$ Because the observations are i.i.d.:

$$P(D|\theta) = \prod_{i=1}^{n} P(x_i|\theta)$$

Using the counts s_1, s_2, s_3:

$$P(D|\theta) = \theta_1^{s_1}(2\theta_1)^{s_2}\theta_2^{s_3}$$

Taking logs:

$$\log(P|\theta) = \log(\theta_1^{s_1}(2\theta_1)^{s_2}\theta_2^{s_3})$$

$$= s_1 \log\theta_1 + s_2\log(2\theta_1) + s_3\log\theta 2$$

Thus: $P(D|\theta) = \theta_1^{s_1}(2\theta_1)^{s_2}\theta_2^{s_3}$,

and: $\log P(D|\theta) = s_1\log\theta_1 + s_2\log(2\theta_1) + s_3\log\theta_2$

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1c We maximize the log-liklihood subject to the constraint

$$3\theta_1 + \theta_2 = 1$$

So write: $\theta_2 = 1 - 3\theta_1$

Sub into log liklihood:

$$\ell(\theta_1) = s_1 \log \theta_1 + s_2 \log(2\theta_1) + s_3 \log(1 - 3\theta_1)$$

Ignoring constant term $s_2 \log 2$;

$$\ell(\theta_1) = (s_1 + s_2) \log \theta_1 + s_3 \log(1 - 3\theta_1) + \text{const}$$

Differentiate w.r.t. θ_1:

$$\frac{d\ell}{d\theta_1} = \frac{s_1 + s_2}{\theta_1} - \frac{3s_3}{1 - 3\theta_1}$$

Set derivative to 0:

$$\frac{s_1 + s_2}{\theta_1} = \frac{3s_3}{1 - 3\theta_1}$$

Cross multiply: $(s_1 + s_2)(1 - 3\theta_1) = 3s_3\theta_1$

Move θ_1 terms together: $s_1 + s_2 = 3\theta_1(s_1 + s_2 + s_3)$

Since: $n = s_1 + s_2 + s_3$

We obtain: $\hat{\theta}_1 = \frac{s_1 + s_2}{3n}$

Then:

$$\hat{\theta}_2 = 1 - 3\hat{\theta}_1$$

$$\hat{\theta}_2 = 1 - \frac{s_1 + s_2}{n}$$

$$\hat{\theta}_2 = \frac{s_3}{n}$$

Thus: $\hat{\theta}_1 = \frac{s_1 + s_2}{3n}$ $\hat{\theta}_2 = \frac{s_3}{n}$

HW4

[2] The likihood is: $L(\beta) = \prod_{i=1}^{n} \frac{1}{\beta} \exp\left(-\frac{x_i}{\beta}\right)$

$$= \beta^{-n} \exp\left(-\frac{1}{\beta}\sum_{i=1}^{n} x_i\right)$$

Take logs: $\ell(\beta) = \log L(\beta)$

$$\ell(\beta) = -n\log\beta - \frac{1}{\beta}\sum_{i=1}^{n} x_i$$

Differentiate w.r.t. β: $\frac{d\ell}{d\beta} = -\frac{n}{\beta} + \frac{1}{\beta^2}\sum_{i=1}^{n} x_i$

Set to 0: $-\frac{n}{\beta} + \frac{1}{\beta^2}\sum_{i=1}^{n} x_i = 0$

Multiply both sides by β^2:

$$-n\beta + \sum_{i=1}^{n} x_i = 0$$

so: $\hat{\beta} = \frac{1}{n}\sum_{i=1}^{n} x_i$

Thus $\hat{\beta} = \bar{x}$

HW4

$\boxed{3a}$ Let D denote the observation of 5 defectives out of 30 items

The likelihood is binomial:

$$p(D|\theta) \propto \theta^5 (1-\theta)^{25}$$

The prior is uniform on $[0,1]$ so:

$$p(\theta) = \propto 1, \quad 0 \leq \theta \leq 1$$

By Bayes rule: $p(\theta|D) \propto p(D|\theta)p(\theta)$

Thus $p(\theta|D) \propto \theta^5(1-\theta)^{25}$ for $0 \leq \theta \leq 1$

Therefore the posterior density up to normalization is

$$p(\theta|D) \propto \theta^5(1-\theta)^{25}, \quad 0 \leq \theta \leq 1$$

This is a beta distribution

$$\theta|D \sim \text{Beta}(6, 26)$$

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HW4

$\boxed{3b}$ Since observations are iid from $N(\mu, 1)$, the liklihood is

$$p(D|\mu) \propto \prod_{i=1}^{n} \exp\left(-\frac{(x_i-\mu)^2}{2}\right)$$

$$\propto \exp\left(-\frac{1}{2}\sum_{i=1}^{n}(x_i-\mu)^2\right)$$

The prior is: $p(\mu) \propto \exp\left(-\frac{\mu^2}{2}\right)$

Therefor, by Bayes rule:

$$p(\mu|D) \propto p(D|\mu)\,p(\mu)$$

$$\propto \exp\left(-\frac{1}{2}\sum_{i=1}^{n}(x_i-\mu)^2\right)\exp\left(-\frac{\mu^2}{2}\right)$$

Combine exponents: $p(\mu|D) \propto \exp\left(-\frac{1}{2}\left[\sum_{i=1}^{n}(x_i-\mu)^2+\mu^2\right]\right)$

Expand the quadratic: $\sum_{i=1}^{n}(x_i-\mu)^2 = \sum_{i=1}^{n}(x_i)^2 - 2\mu\sum_{i=1}^{n}x_i + n\mu^2$

Exponent becomes $-\frac{1}{2}\left[\sum_{i=1}^{n}(x_i)^2 - 2\mu\sum_{i=1}^{n}x_i + n\mu^2 + \mu^2\right]$

$$= -\frac{1}{2}\left[(n+1)\mu^2 - 2\mu\sum_{i=1}^{n}x_i\right] + \text{const}$$

Complete the square: $(n+1)\mu^2 - 2\mu\sum_{i=1}^{n}x_i = (n+1)\left(\mu - \frac{\sum_{i=1}^{n}x_i}{n+1}\right)^2 + \text{const}$

$$= p(\mu|D) \propto \exp\left(-\frac{n+1}{2}\left(\mu - \frac{\sum_{i=1}^{n}x_i}{n+1}\right)^2\right)$$

This is the kernel of a normal distribution with:

$$\text{mean} = \frac{\sum_{i=1}^{n}x_i}{n+1} \qquad \text{variance} = \frac{1}{n+1}$$

Thus: $\mu|D \sim N\left(\frac{\sum_{i=1}^{n}x_i}{n+1}, \frac{1}{n+1}\right)$